



HARSHA
INSTITUTE OF
MANAGEMENT STUDIES

6th SEMESTER
DEPARTMENT

OF

COMPUTER APPLICATIONS

*PRESENTING OUR PROJECT TO ALL
GUIDES AND HOD*

PROJECT TITLE:- RAKSHA SURAKSHA

TEAM MEMBERS

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ABHISHEK

**PROJECT GUIDE:-
JERIN SIR**

**HOD :- NAGA SUMA
MA'AM**

RAKSHA SURAKSHA

Base APP SIZE :- 30 MB to 50 MB

APP INSIDE DOWNLOADABLE SIZE:- PER OFFLINE
zone map 500-700 mb

Our Website access link:-

Rakshasuraksha.github.io

**Our website google search
console:- Raksha Suraksha
Advanced Anti-Theft Safety**



RAKSHA-SURAKSHA

TACTICAL COMMAND SYSTEM

ABSTRACT

- 1) Core Objective: To develop a military-grade personal safety Android application using a 3-Tier security architecture.
- 2) Methodology: The system utilizes background accessibility services, native SmsManager APIs, and offline SQLite Vaults to bypass traditional internet constraints.
- 3) Outcome: A robust mobile command center capable of silently tracking stolen devices, overriding silent profiles for SOS and routing off-grid tactical maps.

System Requirements

HARDWARE

- >Platform: Android Smartphone (Android 8.0 Oreo or higher).
- >Memory (RAM/ROM): Minimum 2GB RAM. App size is ~50MB (ROM), plus ~700MB for the Offline Tactical Map Pack.
- >Core Sensors: GPS Hardware (Tracking), Accelerometer/Gravity Sensor (Crash Detection), Camera (Intruder Capture).
- >Connectivity: Active SIM Card (Mandatory for offline SMS /Telephony overrides).

SOFTWARE

- >Languages: Native Java (Backend Logic) XML (Frontend UI), HTML5/CSS3 (Web Command Hub).
- >Background: AccessibilityService (Fake UI Overlay), BroadcastReceivers (Boot/SIM Detection).
- >Databases: Firebase (Real-time auth, cloud Storage) & SQLite (On-device Offline Evidence vault).
- >Algorithms: Continuous Background Polling (Ghost Tracker) & Vector-based Fall Detection and OSM Offline tiles
- >Dev Environment: AndroidIDE / Android Studio and GithubRepository

APP MODULES

SOS

Auto-senses gravity changes

Instant impact detection

Triggers automated distress message



ANTI-THEFT

Wrong password combo

Pocket removal alarm

Charger disconnect siren

Silent trigger via secret code



TACTICAL RADAR

Offline tracking

Direct satellite GPS pinging

Dead-zone navigation

Multi- zone maps

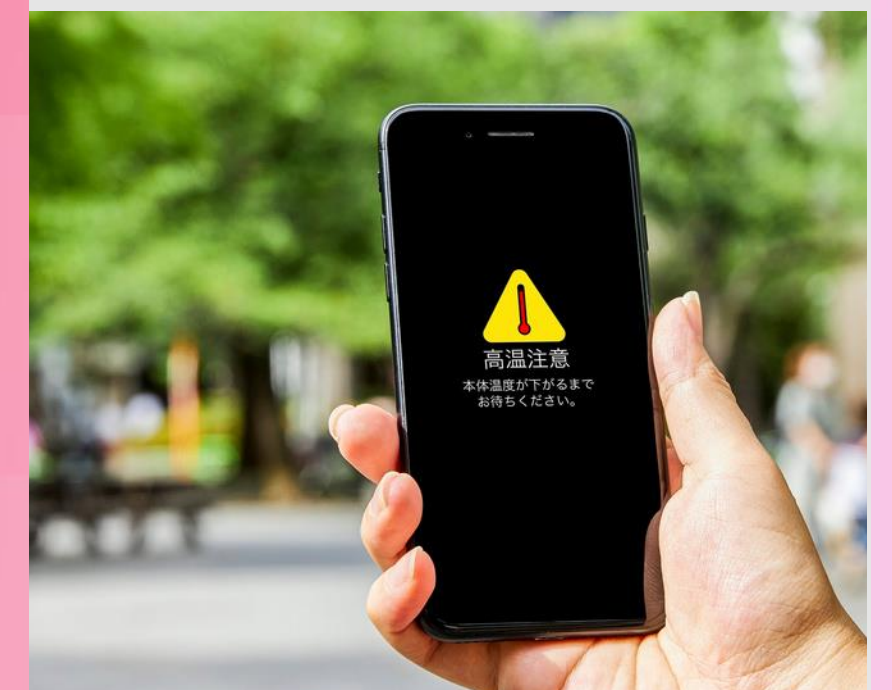


SMART DND OVER RIDE

Auto reply to missed calls

Secret SMS code interception

Forces max-volume alarm bypass



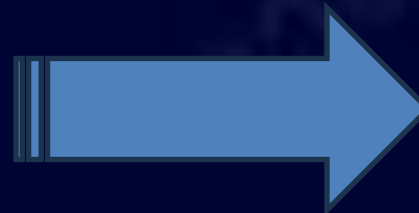
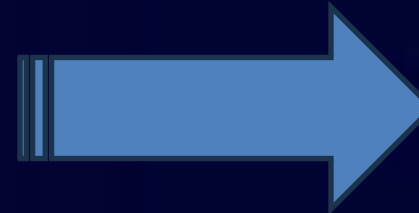
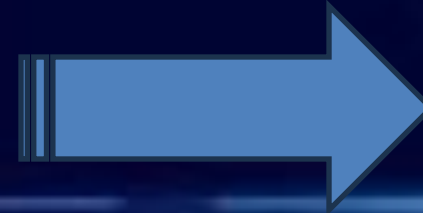
EXISTING Module

Manual SOS Activation
(Requires user to
physically unlock & tap)

Basic Security (Relies only
on lock-screen passwords)

Internet-Dependent
(Tracking fails without
active mobile data)

Static Communication
(Only sends basic, one-
time auto-replies)



OUR Modules

Autonomous SOS Trigger
(Gravity-sensor impact
detection)

Hardware Anti-Theft
(Pocket removal & charger
disconnect alarms)

Offline Tactical Radar
(Direct-satellite tracking
without cell towers)

Active Defence (Remote
SMS override & silent
trigger activation)

APPLICATIONS

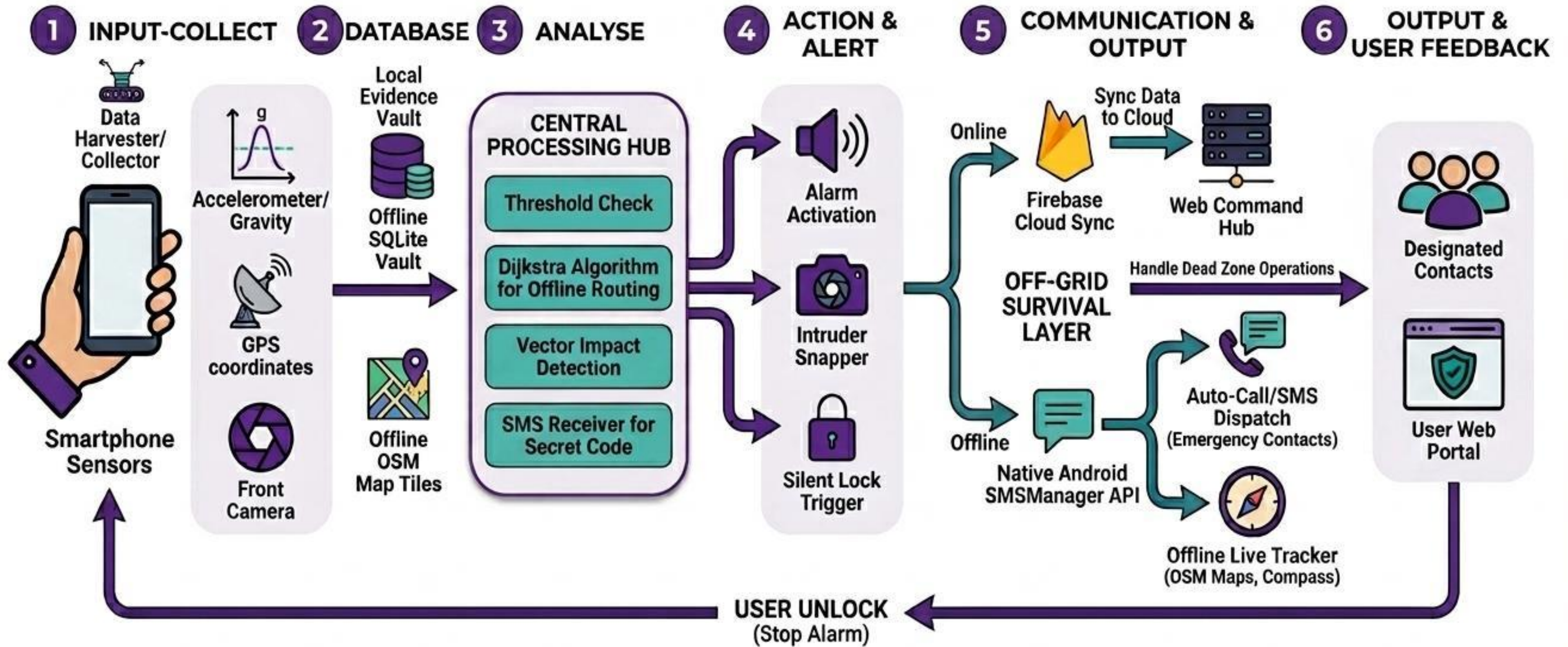
- 1) Personal Safety & Night Commuting:-** Provides instant SOS and auto-calling for women and high-risk transit workers, functioning even in dead zones.
- 2) Vehicle Accident Response:-** Acts as a digital first-responder; g gravity sensors automatically dispatch coordinates if the user is unconscious.
- 3) Anti-Theft & Asset Recovery:-** Tracks stolen devices via remote SMS overrides, SIM-change alerts, and silent intruder logging.
- 4) Off-Grid Adventure & Travel:-** Gives trekkers and remote workers a tactical offline radar for safe routing in zero-connectivity areas.

DATA FLOW

- 1) Input Layer: User manually hits SOS OR Gravity Sensor detects a crash
OR Unauthorized unlock attempt occurs.
- 2) Processing Layer: The Java backend intercepts the trigger.
Decision Branch:
- 3) If Offline: System utilizes Tactical Radar logic, executes remote SMS overrides, and triggers local Auto-Call.
- 4) IfOnline: Pushes data to the Firebase Live Tracker and uploads intruder photos to the Secure Evidence Vault.
- 5) Output Layer: Alerts reach designated contacts and the phone goes into lockdown mode.



DATA FLOW DIAGRAM: RAKSHA SURAKSHA



DRAWBACKS



- 1) Sideloaded & Trust Barrier: Not on Play Store; requires users to manually allow “Unknown Sources.”*
- 2) Complex Uninstallation: “Device Admin” rights block thieves, but make removal harder for actual users.*
- 3) Hardware Override Vulnerability: Software cannot prevent a forced physical shutdown (15-second power hold).*
- 4) Third-Party Quotas: Bound by TRAI regulations (100 SMS/day limit) and Gmail spam filters.*
- 5) High Resource Consumption: Continuous background GPS polling for “Ghost Tracking” drains battery life..*

Future Expectations

- > Direct Emergency Dispatch:** Automated routing of distress calls and exact GPS coordinates directly to local law enforcement, medical services and firefighters services .
- >Zero-Internet Voice Assistant:** A fully offline, localized voice command module to trigger SOS protocols completely hands-free.
- >Wearable Tech Integration:** Synchronizing with smartwatches to allow discreet, one-tap SOS triggering without needing to physically touch the phone.
- >Hardware-Level Integration:** Partnering with mobile manufacturers to embed a dedicated Bluetooth Low Energy (BLE) tracker powered by a reserve battery.
- >Ecosystem Expansion:** Developing a Swift-based version for iOS devices, as for now it is only developed and testing are going for Android operating systems

CONCLUSION

- 1) Active, Not Passive: Upgrades standard, manual SOS buttons into an aggressive, autonomous defense system.
- 2) Offline-First Survival: Guarantees tracking and data rescue in dead zones using native SMS networks.
- 3) Outsmarts Thieves: Deploys psychological traps (like Fake Switch-Off) to defeat physical theft and SIM-swapping.
- 4) The Ultimate Failsafe: Transforms an everyday smartphone into a tactical, off-grid lifeline when standard apps fail.

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